Extension to the LIBOR Market Model (LMM): Calibration to CMS Spread Options & Market Skew Modelling

MRGR Rates

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- The **LIBOR Market Model (LMM)** is a multi-factor model of the joint dynamics of the market-observable forward LIBOR rates. Unlike short-rate models, it prescribes each forward rate's instantaneous volatility and the full inter-rate correlation directly.
- **Tenor structure:** $0 = T_0 < T_1 < \cdots < T_N$, accrual fractions $\tau_n = T_{n+1} - T_n$.
- **Forward rate:**

$$L_n(t) = L_n(t, T_n, T_{n+1}) = \frac{1}{\tau_n}\left(\frac{P(t, T_n)}{P(t, T_{n+1})} - 1\right).$$

- **Dynamics (normal specification):** under the T_{n+1}-forward measure $\mathbb{Q}^{n+1}$:

$$dL_n(t) = \lambda_n(t)^\top dW^{n+1}(t),$$

where $\lambda_n(t) \in \mathbb{R}^m$ is a deterministic instantaneous volatility vector.

- **Decomposition:** the volatility vector $\lambda_n(t)$ is factored as:

$$\lambda_n(t)^\top = \|\lambda_n(t)\| \cdot D_n(t),$$

  where $D_n(t)$ is a vector extracted from the instantaneous correlation matrix of LIBOR rates.
- **Two independent calibration tasks:**
  1. Specify and estimate the instantaneous correlation matrix.
  2. Specify the volatility norm and calibrate it to match market instruments.

- **Volatility norm — grid-based:** $G_{i,j} = \|\lambda_n(t_i)\|$ on an expiry $\times$ tenor grid, with bilinear interpolation between nodes and first-/second-order regularization.
- **Correlation — parametric (Andersen and Piterbarg), with τ_i, τ_j tenors:**

$$r(\tau_i, \tau_j) = \rho_\infty(\tau_{\min}) + \left(1 - \rho_\infty(\tau_{\min})\right) e^{-a(\tau_{\min})\,|\tau_i - \tau_j|}.$$

$$\tau_{\min} = \min(\tau_i, \tau_j),$$

$$a(\tau) = a_\infty + (a_0 - a_\infty)\, e^{-\kappa\tau},$$

$$\rho_\infty(\tau) = b_\infty + (b_0 - b_\infty)\, e^{-\alpha\tau}.$$

- Six parameters $\{a_0, a_\infty, \kappa, b_0, b_\infty, \alpha\}$ fitted to two years of historical EUR forward-rate changes; PCA factor reduction (10 factors, $\sim$97% of variance).

- Following the remark from last time, we check whether the calibrated grid depends on the initial guess: we re-run the regularised swaption calibration from several initialisations.

Initialisation	Final cost	RMS err	Time (ms)
Flat average (baseline)	0.158	0.024	1152
Target-informed	0.158	0.024	876
Random $\pm 50\%$ (5 trials)	0.158	0.024	1150–1330
Target $+$ noise (5 trials)	0.158	0.024	1020–2250

- Every sensible start converges to the *same* optimum (final cost $\approx$ 0.158, RMS $\approx$ 0.024 in normal-vol units): the regularised objective has a single dominant basin, so the reported fit is a global (near-global) minimum and the flat-average default is a sound choice.

- We present here two extensions to the initial framework, developed over the past few months:
- **Extension 1 — Market-implied correlation.** Extend to time-dependent instantaneous correlation, to be calibrated to market-implied correlation via ATM CMS spread option.
- **Extension 2 — Skew.** Extend the deterministic volatility to a *displaced-diffusion* specification, to be calibrated to the skew of swaptions.

- The prior model used a historically estimated stationary correlation, rather than a market-implied term-correlation target.
- It is difficult to determine inter-rate correlation from swaptions alone, since swaption prices mainly constrain volatility levels.
- A CMS spread option with payoff $\max\left(S_1(T) - S_2(T) - K,\ 0\right)$, has a price *highly sensitive to correlation*.

- We generalise the previous correlation structure, making it time-dependent through calibrated nodes.
- *(Exact functional form is proprietary and omitted in this version.)*
- For calibration, a single tenor pair that can be selected by the user (e.g. 10Y-2Y) is used and its CMS spread options over a chosen list of expiries are added.
- **Objective:** swaption fit + CMS spread-option fit + grid smoothness + correlation smoothness.

$$\mathcal{C}(G,\rho) = \frac{w_{sw}}{N_{sw}} \sum_i \left(\bar{\sigma}_{\mathrm{sw},i} - \hat{\sigma}_{\mathrm{sw},i}\right)^2 + \frac{w_{sp}}{N_{sp}} \sum_i \left(\bar{\sigma}_{\mathrm{sp},i} - \hat{\sigma}_{\mathrm{sp},i}\right)^2 + \mathcal{R}_{\mathrm{grid}} + \mathcal{R}_{\rho}.$$

- $\bar{\sigma}$, $\hat{\sigma}$: model / market normal vols (swaption "sw" / spread "sp"); w_{sw}, w_{sp}: fit weights; N_{sw}, N_{sp}: number of swaptions / spread options.
- $\mathcal{R}_{\mathrm{grid}}$, $\mathcal{R}_{\rho}$: first-/second-order smoothness of the vol grid G and of the correlation blend $\rho(t)$ (see Annex).

- How to compute $\bar{\sigma}_{\text{sp}}$ (model-implied spread option volatility)?
- Following Andersen and Piterbarg [2], we can express the spread

$$\varepsilon(T) = S_1(T) - S_2(T) \tag{1}$$

as a Gaussian variable under the T-forward measure.

- Its variance is approximated by

$$\text{Var}^T[\varepsilon(T)] \approx \sum_{i=1}^{2} \int_0^T \|\sigma_{S_i}(t)\|^2 \, \mathrm{d}t - 2\rho_{\text{term}}(0, T) \prod_{i=1}^{2} \left(\int_0^T \|\sigma_{S_i}(t)\|^2 \, \mathrm{d}t \right)^{1/2}, \tag{2}$$

where $\sigma_{S_i}(t)$ denotes the effective instantaneous normal volatility vector of swap rate S_i, obtained from the swaption volatility approximation.

- The term correlation over the interval $[T', T]$ is defined as

$$\begin{aligned} \rho_{\text{term}}(T', T) &= \text{Corr}^T \left(S_1(T) - S_1(T'), S_2(T) - S_2(T') \right) \\ &\approx \frac{\int_{T'}^T \sigma_{S_1}(t)^\top \sigma_{S_2}(t) \, \mathrm{d}t}{\left(\int_{T'}^T \|\sigma_{S_1}(t)\|^2 \, \mathrm{d}t \right)^{1/2} \left(\int_{T'}^T \|\sigma_{S_2}(t)\|^2 \, \mathrm{d}t \right)^{1/2}}. \end{aligned} \tag{3}$$

- The corresponding approximate model-implied normal spread volatility is then obtained from

$$\bar{\sigma}_{\text{sp}} \approx \left(\frac{1}{T} \text{Var}^T[\varepsilon(T)] \right)^{1/2}, \tag{4}$$

- Although the formula we use is slightly more sophisticated, it builds roughly on the same principles. We cannot disclose it due to confidentiality reasons.

- 10Y-2Y CMS spread — ATM normal vols (bps); joint swaption and spread calibration (20 Apr 2026), model results are computed with a Monte Carlo. A good fit to market means that approximations used for CMS spread options are accurate.

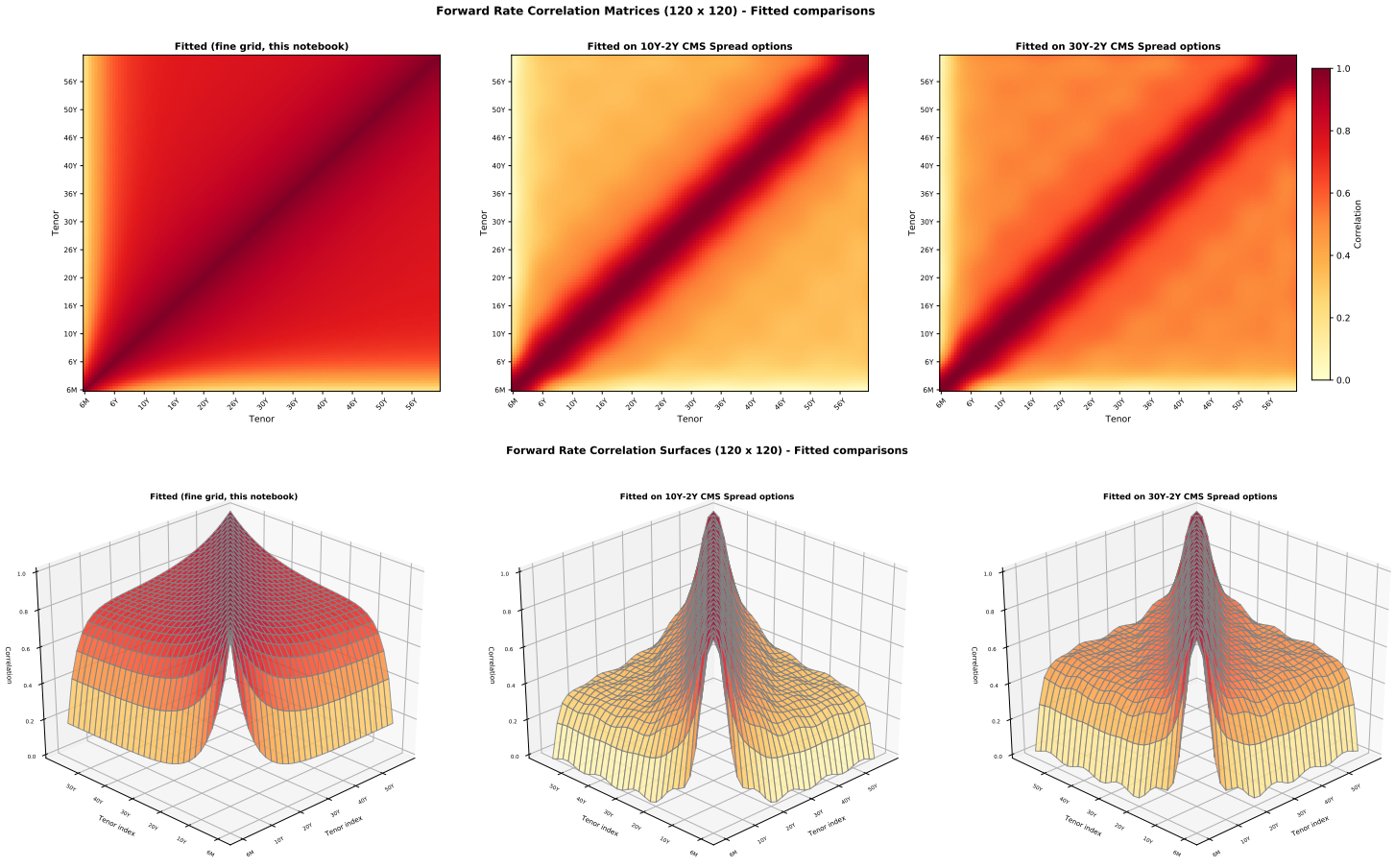
Expiry	Market	Model	Model−Mkt
6M	41.21	41.54	+0.34
1Y	41.64	41.06	−0.58
2Y	39.56	39.27	−0.30
3Y	37.67	37.83	+0.16
4Y	36.33	36.29	−0.04
5Y	35.20	35.11	−0.09
6Y	34.27	34.18	−0.09
7Y	33.46	33.49	+0.03
8Y	32.69	32.55	−0.14
9Y	32.05	31.98	−0.07
10Y	31.63	31.39	−0.24
12Y	30.74	30.74	0.00
15Y	29.78	29.88	+0.10
20Y	28.90	28.97	+0.07
25Y	28.39	28.20	−0.19
30Y	28.10	27.65	−0.46

- MC reproduces market within ±0.6 bps; approximation within ±0.42 bps of MC. Time to calibrate: 17.24 s (vs. 948 ms swaption-only). Swaption fit preserved (RMSE 0.54 bps).

- Results for 30Y-2Y CMS spread.

Expiry	Market	Model	Model−Mkt
6M	61.02	62.21	+1.19
1Y	60.44	60.78	+0.34
2Y	57.12	57.03	−0.10
3Y	54.24	54.69	+0.45
4Y	52.14	52.36	+0.22
5Y	50.30	50.58	+0.28
6Y	48.81	48.66	−0.15
7Y	47.51	46.95	−0.55
8Y	46.31	45.61	−0.70
9Y	45.34	44.71	−0.63
10Y	44.63	43.86	−0.76
12Y	43.21	42.33	−0.88
15Y	41.72	40.98	−0.74
20Y	40.22	39.57	−0.65
25Y	40.33	39.02	−1.31
30Y	40.67	38.92	−1.75

- MC within $\sim \pm 1.75$ bps of market. Time to calibrate: 27.73 s.



- The heatmaps (top) and 3D surfaces (bottom) compare the historical fit (left) with the 10Y-2Y (middle) and 30Y-2Y (right) calibrations: the average correlation between rates 20 years apart drops from 0.76 (historical) to 0.50 (30Y-2Y) and 0.38 (10Y-2Y).

- The normal LMM does not have the capability to fit the market implied skew of swaptions.
- We thus introduce a displaced diffusion. The displaced lognormal model allows a more flexible model, and allows us to calibrate to the smile of swaptions through a displacement surface fitted to the market-implied skew.

- Same tenor structure $0 = T_0 < \cdots < T_N$ and forwards $L_n(t) = \frac{1}{\tau_n}\left(\frac{P(t,T_n)}{P(t,T_{n+1})} - 1\right)$.
- The *normal* diffusion is replaced by a *displaced-diffusion* local volatility. Dynamics under the T_{n+1}-forward measure $\mathbb{Q}^{n+1}$:

$$dL_n(t) = \Big(b_n(t)L_n(t) + \big(1 - b_n(t)\big)L_n(0)\Big)\,\lambda_n(t)^\top dW^{n+1}(t),$$

- The displacement $b_n(t)$ carries the skew: $b_n = 0$ recovers the normal LMM, $b_n = 1$ is lognormal.
- $b_n(t)$ is parameterised on a piecewise-linear upper triangular grid.

- **Objective:** previous cost function extended to include swaptions on the smile (3 points per pair (expiry, tenor)) and implied swap rate displacement b_S from market smile and smoothing of the LIBOR displacement grid $b_n(t)$.

$$\mathcal{C}_{\mathrm{DLN}}(G, b, \rho) = \frac{w_{sw}}{N_{sw}} \sum_i \left(\bar{\sigma}^{\mathrm{DLN}}_{\mathrm{sw},i} - \hat{\sigma}^{\mathrm{DLN}}_{\mathrm{sw},i}\right)^2 + \frac{w_{sp}}{N_{sp}} \sum_i \left(\bar{\sigma}_{\mathrm{sp},i} - \hat{\sigma}_{\mathrm{sp},i}\right)^2$$
$$+ \frac{w_b}{N_b} \sum_i \left(\bar{b}_{S,i} - \hat{b}_{S,i}\right)^2 + \mathcal{R}_{\mathrm{grid}} + \mathcal{R}_b + \mathcal{R}_\rho .$$

- **Arguments** — G: instantaneous vol-norm grid; b: LIBOR displacement grid; ρ: time-dependent correlation.
- **Swaption smile** — $\bar{\sigma}^{\mathrm{DLN}}_{\mathrm{sw}}$, $\hat{\sigma}^{\mathrm{DLN}}_{\mathrm{sw}}$: model / market displaced-diffusion swaption vols; w_{sw}, N_{sw}: fit weight / number (3 strikes per expiry–tenor).
- **Spread** — $\bar{\sigma}_{\mathrm{sp}}$, $\hat{\sigma}_{\mathrm{sp}}$: model / market normal CMS-spread vols; w_{sp}, N_{sp}: fit weight / number.
- **Displacement** — $\bar{b}_S$, $\hat{b}_S$: model- / market-implied displacement of swap rates; w_b, N_b: fit weight / number.
- **Regularisation** — $\mathcal{R}_{\mathrm{grid}}$ (vol grid), $\mathcal{R}_b$ (displacement), $\mathcal{R}_\rho$ (correlation): 1st/2nd-order smoothing (see annex).

- How to compute $\hat{\sigma}_{\text{sw}}^{\text{DLN}}$ (market-implied displaced diffusion swaption volatilities) and $\hat{b}_S$ (market-implied swap rate displacement)?
- We model each swap rate as a displaced diffusion with constant coefficients under its annuity measure $\mathbb{Q}^A$:

$$dS(t) = \left(b_S\, S(t) + (1 - b_S)\, S(0)\right) \sigma_S\, dW^A(t).$$

- We price the corresponding swaptions with a shifted Black formula, and invert with Bachelier to a normal vol $\bar{\sigma}^{\text{normal}}(\sigma_S, b_S)$:
- For each swaption we then fit the market smile to obtain suitable targets for the cost function:

$$(\hat{\sigma}_{\text{sw}}^{\text{DLN}}, \hat{b}_S) = \underset{\sigma_S, b_S}{\arg\min} \left(\bar{\sigma}^{\text{normal}}(\sigma_S, b_S) - \hat{\sigma}_{\text{swaption}}^{\text{normal}}\right)^2.$$

- How to compute $\bar{\sigma}_{\text{sw}}^{\text{DLN}}$ (model-implied displaced diffusion swaption volatilities) and $\bar{b}_S$ (model-implied swap rate displacement)?
- Following Andersen and Piterbarg [2], for a swap making payments at times $T_{j+1}, \ldots, T_k$, under a given calibrated LMM, its swap rate dynamics under the annuity measure are given by:

$$dS(t) = \Big(b_S(t)\, S(t) + (1 - b_S(t))\, S(0) \Big)\, \sigma_S(t)\, dW^A(t),$$

  where $\sigma_S(t)$ and $b_S(t)$ are time-dependent functions dependent on $\lambda_n(t)$ and $b_n(t)$ evaluated at times $T_j, \ldots, T_{k-1}$.
- Through parameter averaging, we can freeze the time-dependent coefficients to obtain the following swap rate dynamics:

$$dS(t) = \Big(b_S^{\text{eff}}\, S(t) + (1 - b_S^{\text{eff}})\, S(0) \Big)\, \sigma_S^{\text{eff}}\, dW^A(t),$$

  where σ_S^{eff} and b_S^{eff} are effective (time-averaged) coefficients determined by the calibrated LMM parameters.
- We then take σ_S^{eff} as $\bar{\sigma}_{\text{sw}}^{\text{DLN}}$ and b_S^{eff} as $\bar{b}_S$.

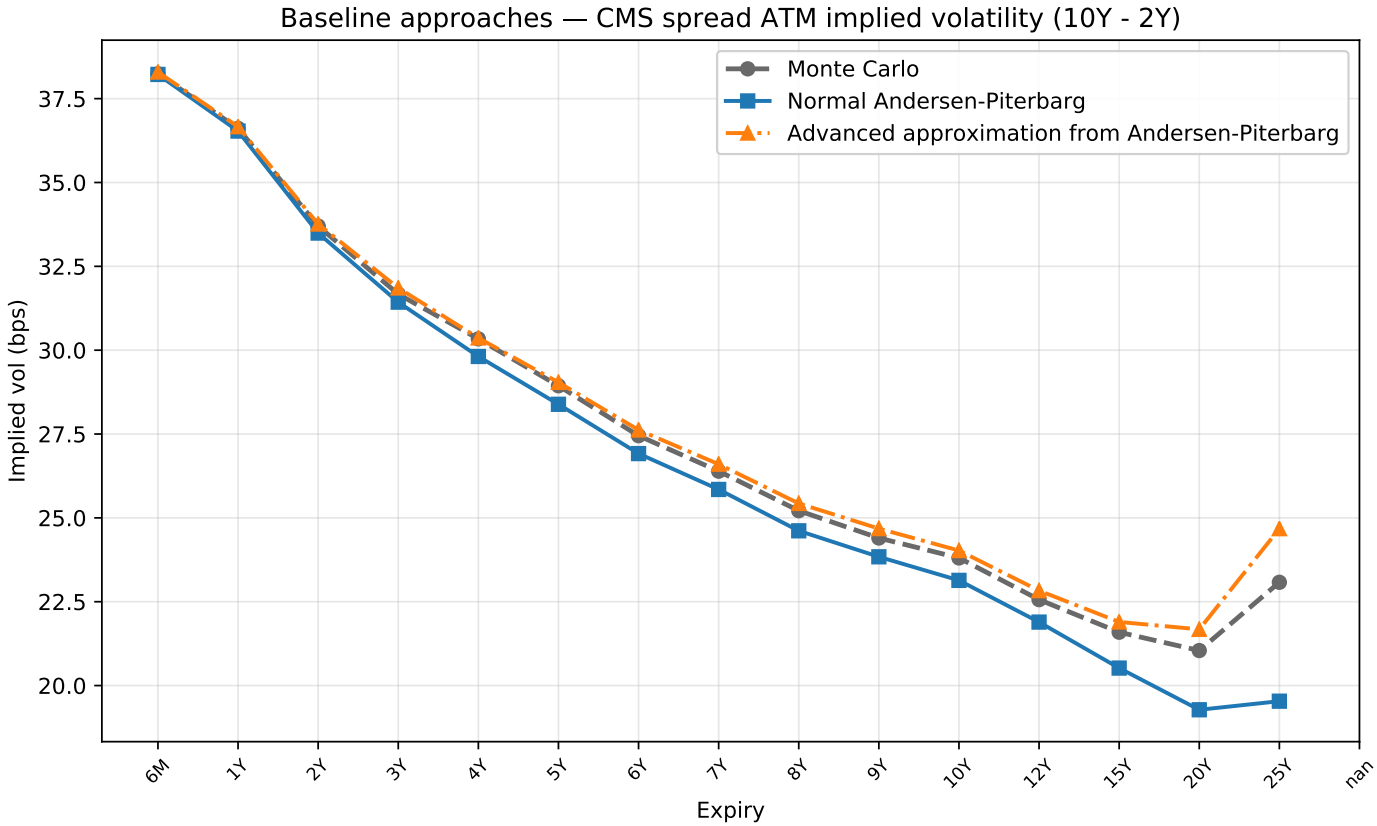
- How to compute $\bar{\sigma}_{\text{sp}}$ (model-implied CMS spread option volatility)?
- Approximating the CMS-spread option volatility under the displaced-diffusion LMM proved substantially harder. We first present the method exposed in Andersen and Piterbarg [2].
- Following [2], the undiscounted CMS spread option (maturity T, strike K, legs S_1, S_2) can be priced with a two-dimensional Gaussian-copula integral over the two swap-rate marginals:

$$V_{CMS}(0) \approx \iint \left((\Psi_1^T)^{-1}(\Phi(z_1)) - (\Psi_2^T)^{-1}(\Phi(z_2)) - K \right)^+ \phi\left(z_1, z_2; \rho_{\text{term}}\right) dz_1\, dz_2,$$

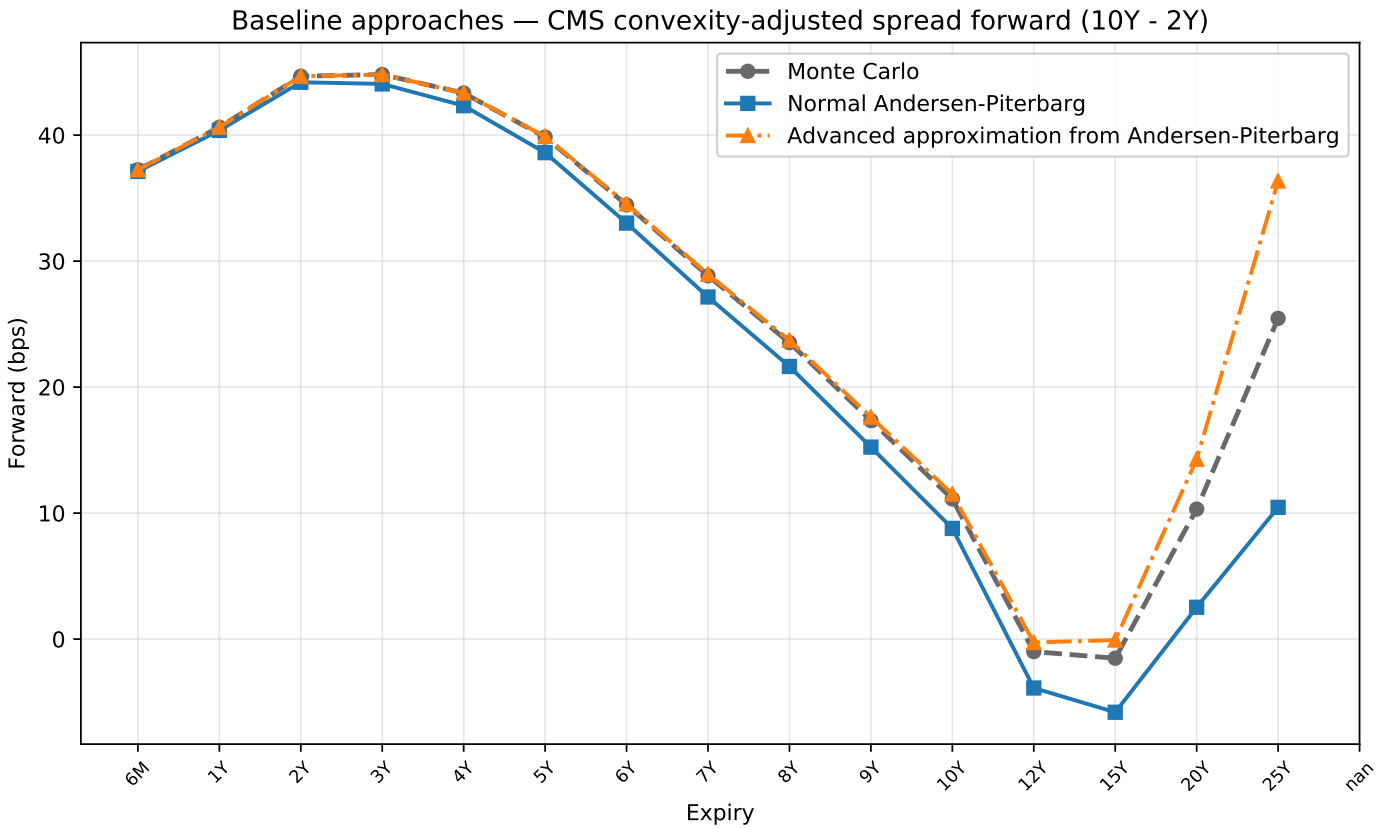
$$\Psi^T(s) = \frac{A(0)}{P(0,T)} \int_{-\infty}^{s} \alpha(u)\, \psi^A(u)\, du,$$

$$\alpha(s) = E^A\left[\frac{1}{A(T)} \,\middle|\, S(T) = s \right] \quad \text{(annuity mapping)}.$$

- Φ: standard normal CDF; $\phi(\cdot; \rho_{\text{term}})$: bivariate Gaussian density. Each marginal CDF Ψ_i^T is rebuilt from the annuity mapping α and the conditional expectation of the vector of LIBORs given the swap rate.
- In addition to this method, Andersen and Piterbarg suggest a simpler approximation that does not take displacement into account, we will present the results obtained by using both methods in the two following slides.

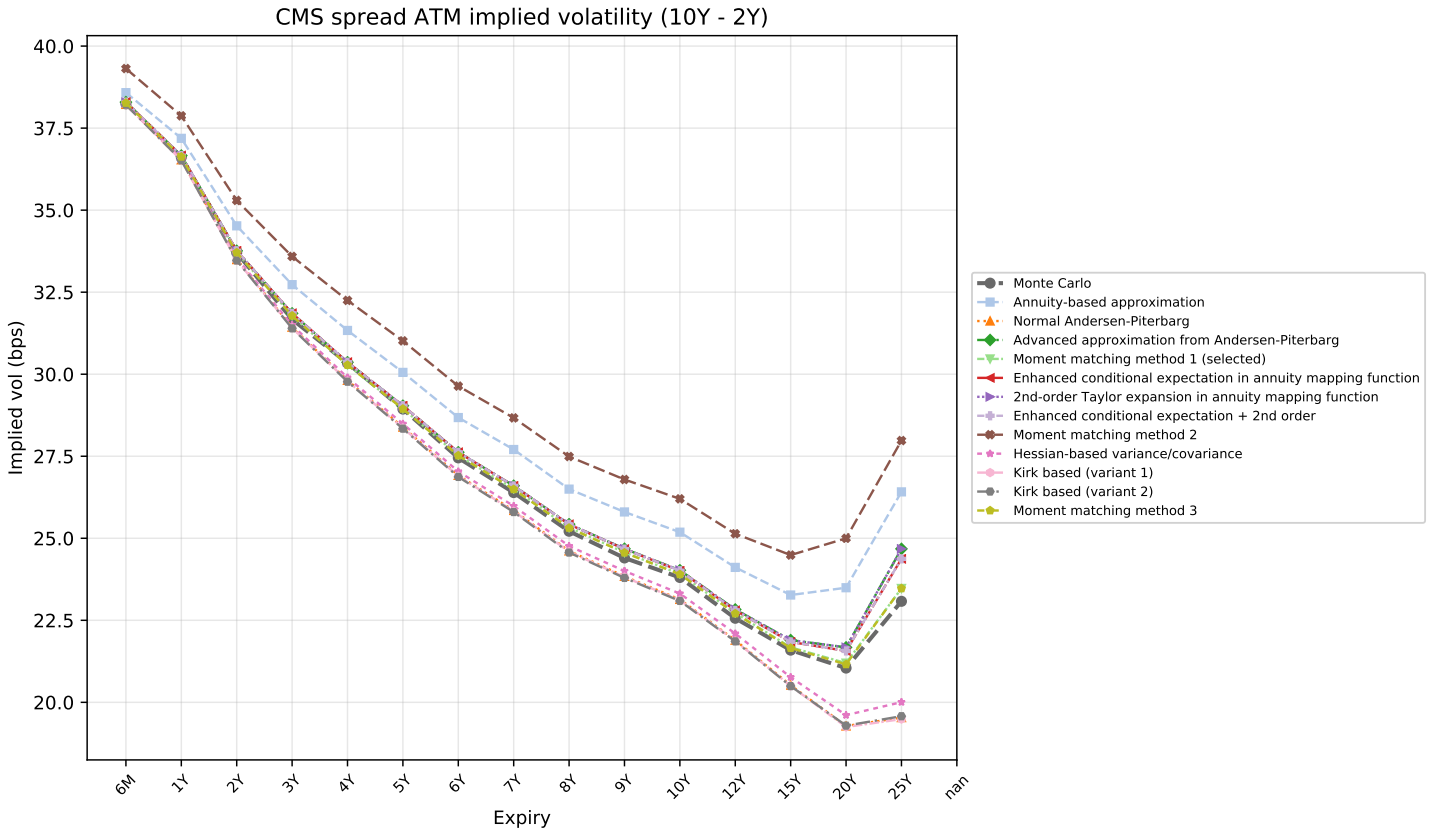


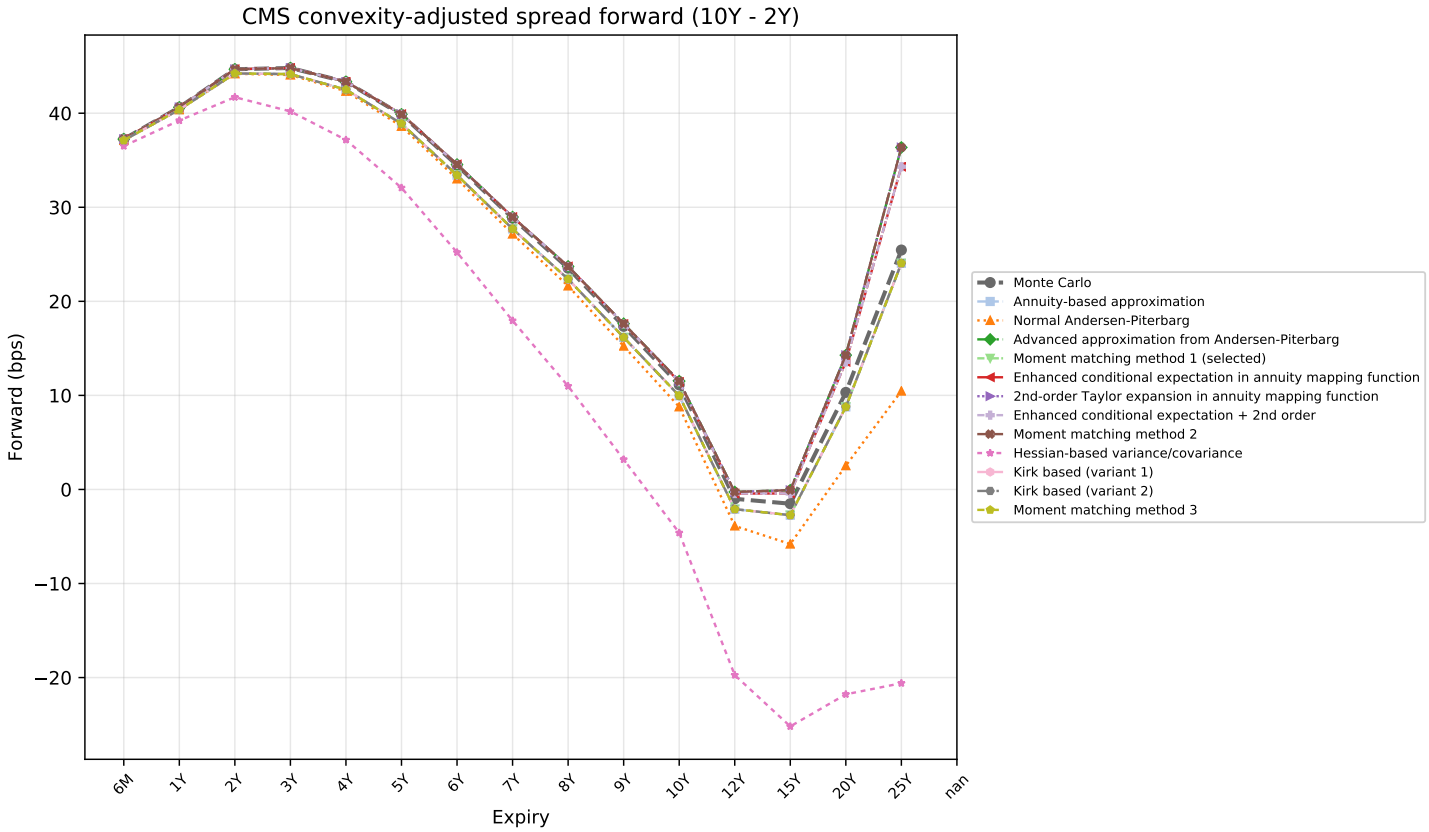
- 10Y-2Y CMS spread, ATM implied normal vol (bps) vs. expiry: Monte-Carlo (grey) against the two baseline approximations.



- 10Y-2Y CMS spread, convexity-adjusted (CMS) forward vs. expiry.

- We therefore tested more than 15 alternative approximations. 10Y-2Y ATM implied normal vol (bps) vs. expiry: Monte-Carlo (grey) against the full family of candidates — several track Monte-Carlo more closely than the method presented in Andersen and Piterbarg.





- Convexity-adjusted (CMS) forward vs. expiry for the 10Y-2Y basket, same method family. The retained approach (next section) is selected from these candidates on accuracy against Monte-Carlo across expiries.

- Displaced-lognormal model calibrated jointly to the swaption smile (3 strikes) + CMS spreads; residuals of the Monte-Carlo model (bps).

	Model − Market	
Instrument	RMSE	Max
Swaptions, ATM−25 bp	0.66	3.07
Swaptions, ATM	0.52	1.48
Swaptions, ATM+25 bp	0.76	3.44
CMS 30Y-2Y	1.40	2.18
CMS 10Y-2Y	0.30	0.61

- Swaption smile matched within $<$ 1 bps RMSE; CMS spreads accurate (10Y-2Y within 0.6 bps); the wider 30Y-2Y shows a long-expiry bias ($\leq$ 2.2 bps).
- **Cost:** full displaced-lognormal calibration (smile and CMS spreads) in 96.14 s, vs. 27.73 s correlation-only and 948 ms swaption-only.

- Per-expiry Monte-Carlo residuals (bps) vs. market (Model−Mkt) and vs. the closed-form approximation (Model−Apx), with the MC standard error (s.e.).

30Y-2Y

Exp.	Model−Mkt	Model−Apx	s.e.
6M	−0.07	−0.21	0.26
1Y	−0.47	−0.41	0.26
2Y	−0.59	−0.59	0.25
3Y	−0.76	−0.80	0.23
4Y	−0.62	−0.64	0.22
5Y	−0.73	−0.79	0.21
6Y	−0.96	−0.92	0.21
7Y	−1.11	−1.18	0.20
8Y	−1.39	−1.36	0.19
9Y	−1.53	−1.60	0.19
10Y	−1.67	−1.67	0.18
12Y	−1.93	−1.95	0.17
15Y	−2.18	−2.19	0.16
20Y	−2.13	−2.15	0.16
25Y	−2.01	−2.02	0.17
30Y	−1.74	−1.74	0.19

10Y-2Y

Exp.	Model−Mkt	Model−Apx	s.e.
6M	+0.45	−0.11	0.19
1Y	−0.61	−0.14	0.19
2Y	−0.32	−0.25	0.18
3Y	−0.16	−0.30	0.17
4Y	−0.30	−0.34	0.17
5Y	−0.17	−0.22	0.16
6Y	−0.15	−0.08	0.16
7Y	+0.02	−0.11	0.15
8Y	−0.23	−0.27	0.15
9Y	−0.21	−0.34	0.15
10Y	−0.25	−0.23	0.14
12Y	−0.10	−0.08	0.14
15Y	−0.18	−0.19	0.13
20Y	−0.34	−0.33	0.13
25Y	−0.35	−0.38	0.13
30Y	−0.43	−0.38	0.14

Residuals stay within ~ 2 bps; the 10Y-2Y fit sits within a few MC standard errors, confirming a tight joint fit.

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	−1.93	−0.10	−3.07	−1.10	−0.78	−0.60	−0.06	+0.50	−0.18	−0.53	−1.32	−1.34
1Y	−1.13	+0.66	−0.84	−0.20	−0.47	−0.82	−0.79	−0.84	−1.11	−1.17	−1.43	−1.09
2Y	−0.23	+1.19	+0.17	+0.12	−0.21	−0.35	−0.02	−0.23	−0.28	−0.40	−0.71	−0.79
3Y	−0.43	+0.86	+0.03	+0.05	+0.06	−0.31	−0.03	−0.38	−0.34	−0.35	−0.57	−0.95
4Y	−0.30	+1.06	+0.19	+0.41	+0.31	−0.10	−0.02	−0.29	−0.40	−0.49	−0.86	−0.83
5Y	−0.52	+0.98	+0.24	+0.27	+0.17	−0.26	−0.21	−0.19	−0.30	−0.33	−0.72	−0.52
7Y	−0.19	+1.09	+0.23	+0.09	−0.06	−0.06	+0.24	−0.13	−0.13	−0.10	−0.51	−0.68
10Y	−0.46	+1.08	−0.28	−0.24	+0.16	+0.07	+0.12	+0.10	+0.07	+0.01	−0.54	−0.69
12Y	−0.62	+1.09	+0.28	+0.13	+0.01	−0.10	+0.16	+0.12	+0.04	+0.01	−0.35	−0.76
15Y	−0.19	+1.28	+0.18	+0.33	+0.41	+0.28	+0.37	+0.08	−0.08	−0.17	−0.36	−1.01
20Y	−0.33	+1.08	−0.05	+0.01	+0.13	+0.04	+0.26	−0.10	−0.27	−0.36	−0.67	−1.38
30Y	+0.32	+1.67	+0.55	+0.47	+0.42	+0.74	+0.71	+0.13	−0.19	−0.13	−0.61	−1.26

Figure: Swaption residuals: Monte-Carlo model − market normal volatility (bps), strike offset ATM−25 bp. Rows: expiry, columns: tenor (RMSE = 0.66 bps, max = 3.07 bps).

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	+0.41	+1.30	−1.26	+0.32	+0.45	+0.49	+0.73	+0.67	+0.27	+0.31	+0.31	+1.02
1Y	+0.06	+1.44	+0.47	+1.06	+0.81	+0.52	+0.64	+0.28	+0.07	+0.09	+0.13	+0.71
2Y	0.00	+1.34	+0.86	+0.75	+0.40	+0.27	+0.42	−0.05	+0.02	+0.09	+0.16	+0.29
3Y	−0.37	+0.85	+0.36	+0.36	+0.37	+0.02	+0.14	−0.40	−0.27	−0.09	+0.03	−0.17
4Y	−0.35	+0.97	+0.36	+0.59	+0.53	+0.16	+0.12	−0.27	−0.28	−0.19	−0.28	−0.14
5Y	−0.59	+0.89	+0.37	+0.42	+0.37	−0.01	−0.09	−0.22	−0.23	−0.09	−0.23	0.00
7Y	−0.33	+0.97	+0.22	+0.12	+0.02	+0.08	+0.32	−0.18	−0.10	+0.02	−0.22	−0.40
10Y	−0.50	+1.07	−0.30	−0.22	+0.20	+0.14	+0.11	+0.03	+0.04	+0.06	−0.42	−0.64
12Y	−0.67	+1.05	+0.23	+0.12	+0.03	−0.04	+0.16	+0.06	+0.02	+0.02	−0.29	−0.74
15Y	−0.35	+1.16	+0.07	+0.26	+0.38	+0.30	+0.35	−0.01	−0.15	−0.22	−0.37	−1.07
20Y	−0.41	+1.04	−0.11	−0.02	+0.12	+0.05	+0.19	−0.28	−0.44	−0.51	−0.76	−1.48
30Y	−0.13	+1.26	+0.15	+0.10	+0.10	+0.44	+0.40	−0.19	−0.47	−0.36	−0.75	−1.40

Figure: Swaption residuals: Monte-Carlo model − market normal volatility (bps), strike offset ATM. Rows: expiry, columns: tenor (RMSE = 0.52 bps, max = 1.48 bps).

Expiry \ Tenor	6M	1Y	2Y	3Y	4Y	5Y	7Y	10Y	12Y	15Y	20Y	30Y
6M	−1.41	−1.37	−3.44	−2.07	−1.95	−1.81	−1.76	−2.25	−2.50	−2.18	−1.58	−0.37
1Y	−0.80	+0.25	−0.17	+0.37	+0.20	+0.06	+0.32	−0.30	−0.49	−0.40	−0.08	+0.58
2Y	−0.72	+0.56	+0.57	+0.41	+0.05	−0.05	−0.06	−0.82	−0.63	−0.39	−0.03	+0.34
3Y	−0.93	+0.24	+0.08	+0.06	+0.09	−0.23	−0.27	−1.02	−0.79	−0.46	−0.03	−0.03
4Y	−0.83	+0.44	+0.10	+0.34	+0.30	−0.04	−0.21	−0.72	−0.60	−0.37	−0.20	+0.07
5Y	−1.03	+0.45	+0.16	+0.24	+0.21	−0.13	−0.32	−0.59	−0.52	−0.26	−0.16	+0.15
7Y	−0.68	+0.62	0.00	−0.07	−0.13	−0.03	+0.13	−0.47	−0.33	−0.10	−0.20	−0.37
10Y	−0.66	+0.92	−0.46	−0.34	+0.11	+0.06	−0.05	−0.21	−0.15	−0.09	−0.50	−0.77
12Y	−0.82	+0.92	+0.08	−0.01	−0.07	−0.11	+0.02	−0.14	−0.16	−0.11	−0.38	−0.88
15Y	−0.59	+0.95	−0.14	+0.10	+0.25	+0.22	+0.22	−0.21	−0.33	−0.38	−0.50	−1.24
20Y	−0.57	+0.91	−0.23	−0.13	+0.03	−0.01	+0.05	−0.54	−0.71	−0.74	−0.96	−1.66
30Y	−0.62	+0.80	−0.30	−0.31	−0.29	+0.10	+0.03	−0.58	−0.81	−0.65	−0.97	−1.59

Figure: Swaption residuals: Monte-Carlo model − market normal volatility (bps), strike offset ATM+25 bp. Rows: expiry, columns: tenor (RMSE = 0.76 bps, max = 3.44 bps).

- **Extensions of the normal LMM added in the internal benchmarking library.** During the past few months, we successfully integrated a LIBOR Market Model calibrated to spread options and to the skew of swaptions.
  - **Accuracy on spread options in the normal case.** In the normal case, we achieve fitting RMSEs of 0.24 bps on 10Y-2Y ATM spread options and 0.80 bps on 30Y-2Y ATM spread options.
  - **Skew.** For the skew, the model achieves a 0.5–0.8 bps RMSE on swaptions on the smile (3 strikes per expiry–tenor) and 0.30 bps (10Y-2Y) / 1.40 bps (30Y-2Y) RMSE on ATM spread options.
  - **Performance** (single-threaded, Intel Xeon Gold 6240R). Regularised swaption-only 948 ms; joint swaption + CMS spread 17.24 s (10Y-2Y) / 27.73 s (30Y-2Y); full displaced-lognormal (smile + spreads + skew) 96.14 s.
  - **Integration to internal library.** Each component of those extensions has been integrated to the internal benchmarking library with necessary unit tests and regression tests and provided with a user-friendly interface and several example spreadsheets to demonstrate usage to users of the library.

- **Limitations:**
  - Displaced diffusion captures the skew but not the full smile curvature (the wings would need a stochastic-volatility overlay);
- **Current and next steps:**
  - Add a stochastic-volatility overlay on the displaced-diffusion skew (CEV / stochastic volatility) for the smile wings;

- **Option 1 — Parametric (Rebonato [27]):**
  - Time-stationary: $g(t, x) = g(x)$; or separable: $g(t, x) = g_1(t)\, g_2(x)$.
  - Few parameters $\Rightarrow$ stable calibration, but limited flexibility to match the full surface.
  - Example: $\lambda(T_n - t) = (a + b(T_n - t))\, e^{-c(T_n - t)} + d$, with $T_n - t$ being time to reset — captures empirical hump shape.
- **Option 2 — Grid-based ([25], [32], [2]) (our choice):**
  - Rectangular grid $G_{i,j} = \|\lambda_n(t_i)\|$ over expiries t_i and tenors x_j, with bilinear interpolation between nodes.
  - $N_t \times N_x$ degrees of freedom $\Rightarrow$ can match the entire swaption/caplet surface.
  - Requires regularization to prevent overfitting.
  - Current industry standard for global calibration.

- The instantaneous correlation of the forward-rate increments fixes the *direction* of the volatility vectors; their norm fixes the *length*:

$$\mathrm{Corr}\big(dL_j(t), dL_k(t)\big) = \frac{\lambda_j(t)^\top \lambda_k(t)}{\|\lambda_j(t)\|\,\|\lambda_k(t)\|}.$$

- We therefore write each m-factor volatility vector as its norm times a *unit direction* $D_n(t) \in \mathbb{R}^m$ taken from the correlation structure:

$$\lambda_n(t)^\top = \|\lambda_n(t)\|\, D_n(t), \qquad \|D_n(t)\| = 1, \qquad \|\lambda_n(t)\| = G(t, T_n - t).$$

- The directions satisfy $D_j(t) \cdot D_k(t) = \mathrm{Corr}(dL_j, dL_k)$, so the target correlation is reproduced exactly and $\|\lambda_n(t)\|$ is the marginal (normal/Bachelier) instantaneous volatility, read from the calibrated vol-norm grid G.
- Calibration then splits cleanly: (i) correlation $\Rightarrow$ $D_n(t)$ (parametric kernel + PCA, next slide); (ii) norm $\Rightarrow$ G, fitted to swaptions.

- Assuming correlation depends only on tenor differences, the $p(t)$ forwards still alive at t ($L_{q(t)}, \ldots, L_{N-1}$, with $q(t)$ the first-alive index) have correlation matrix built from the parametric kernel r:

$$\left(R_{q(t)}\right)_{ij} = r\left(T_{q(t)+i-1} - T_{q(t)},\ T_{q(t)+j-1} - T_{q(t)}\right), \qquad i, j = 1, \ldots, p(t).$$

- **Correlation PCA** (Andersen–Piterbarg §14.3.4): choose a diagonal shift $\mu \in \mathbb{R}^{p(t)}$ so that the rank-m eigen-truncation of $R_{q(t)} + \mathrm{diag}(\mu)$ keeps a *unit diagonal*:

$$R_{q(t)} + \mathrm{diag}(\mu) = E \Lambda E^{\top}, \qquad D_{q(t)} = E_{*,1:m}\, \Lambda_{1:m}^{1/2}, \qquad \left(D_{q(t)} D_{q(t)}^{\top}\right)_{ii} = 1 \ \ \forall i.$$

- Then $R_{q(t)} \approx D_{q(t)} D_{q(t)}^{\top}$ is a valid rank-m correlation, and $D_n(t)$ is its $\left(n - q(t) + 1\right)$-th row. The shift μ is found by Levenberg–Marquardt on the diagonal residual $\left(D_{q(t)} D_{q(t)}^{\top}\right)_{ii} - 1$.

- **Rebonato frozen-weight approximation.** For a swap spanning T_j to T_k, the effective swap-rate volatility vector is a frozen-weight sum of the LIBOR volatilities, and its squared norm is a *double sum* of LIBOR volatilities:

$$\lambda_S(t) \approx \sum_{n=j}^{k-1} w_n(0)\,\lambda_n(t), \qquad w_n(0) = \frac{\partial S(0)}{\partial L_n(0)},$$

$$\|\lambda_S(T_n)\|^2 = \sum_{l=j}^{k-1}\sum_{m=j}^{k-1} w_l(0)\,w_m(0)\;\lambda_l(T_n)^\top \lambda_m(T_n).$$

- The scalar swap-rate normal volatility σ (used on the CMS-convexity slide) is its expiry time-average:

$$\sigma \equiv \bar{\sigma}_{\text{swaption}} = \left(\frac{1}{T_j}\int_0^{T_j} \|\lambda_S(t)\|^2\,dt\right)^{1/2} \approx \left(\frac{1}{T_j}\sum_{n=1}^{j} \tau_{n-1}\,\|\lambda_S(T_n)\|^2\right)^{1/2}.$$

- $\lambda_n(t)$: instantaneous LIBOR volatility vector; $w_n(0)$: frozen annuity weight; $\tau_{n-1} = T_n - T_{n-1}$.

- Smoothing penalties in the objectives (first-/second-order finite differences on the grids):

$$\mathcal{R}_{\text{grid}}(G) = \frac{1}{N_t N_x} \sum_{i,j} \Big(\lambda_t \|\partial_t G\|^2 + \lambda_x \|\partial_x G\|^2 + \lambda_{tt} \|\partial_t^2 G\|^2 + \lambda_{xx} \|\partial_x^2 G\|^2 \Big),$$

$$\mathcal{R}_\rho(\rho) = \frac{\lambda_{\rho,t}}{N_{sp}} \|\partial_t \rho\|^2 + \frac{\lambda_{\rho,tt}}{N_{sp}} \|\partial_t^2 \rho\|^2,$$

$$\mathcal{R}_b(b) = \frac{1}{N_t N_x} \sum_{i,j} \left(\lambda_{b,1} \|\nabla b\|^2 + \lambda_{b,2} \|\nabla^2 b\|^2 \right).$$

- λ_t, λ_x (1st-order) / $\lambda_{tt}, \lambda_{xx}$ (2nd-order): grid weights; $\lambda_{\rho,\bullet}$: correlation weights; $\lambda_{b,\bullet}$: displacement weights.

- **Problem structure.** The calibration objective is a nonlinear least-squares sum of M residuals in P parameters, $\mathcal{C}(\theta) = \|\mathbf{r}(\theta)\|^2$ (weighted fit + regularization), minimised over θ.
- **LM update.** With $J = \partial \mathbf{r}/\partial\theta$ the $M \times P$ residual Jacobian, the step damps the Gauss–Newton normal equations by $\mu \geq 0$:

$$\theta_{k+1} = \theta_k - \left(J^\top J + \mu\, I\right)^{-1} J^\top \mathbf{r}(\theta_k).$$

- **Adaptive damping.** μ is retuned each step — shrunk after a cost-reducing step, grown after a rejected one (a self-tuning trust region, no line search) — and interpolates two regimes:
  - $\mu \to 0$: Gauss–Newton — fast (locally quadratic) near the optimum;
  - $\mu \to \infty$: (scaled) gradient descent — stable far from it.
- **Why LM over L-BFGS?** LM exploits the full $M \times P$ residual sensitivities each iteration; quasi-Newton (L-BFGS) sees only the P aggregated gradient components. On this least-squares problem LM reaches the same optimum in 1–2 orders of magnitude less wall-clock time (Fig. 4).

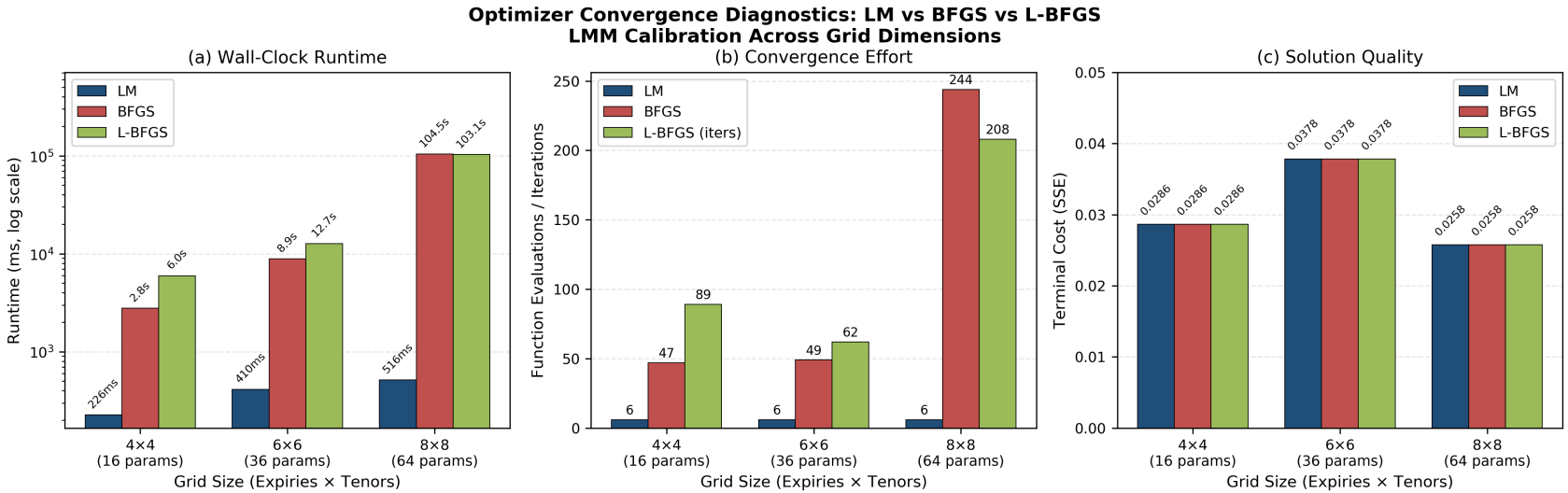


Figure: Optimizer convergence: (a) wall-clock runtime (log scale); (b) function evaluations; (c) terminal cost. LM, L-BFGS and BFGS converge to an *identical* objective, but LM attains it in 1–2 orders of magnitude less time.

- More than fifteen approximations were tested against Monte Carlo, we roughly present the core idea of each.

Approach	**Core idea**
Closed-form Gaussian (moment) spread vol	
Normal Andersen–Piterbarg	Per-leg normal vols; no convexity adjustment.
Annuity-based approximation	DD moments + per-leg CMS convexity adjustment.
Annuity mapping + 1-D Gaussian copula	
Advanced approximation (A–P)	Affine annuity-mapping DD marginals; copula integral.
Enhanced conditional expectation	As above; Better conditional-Libor curve.
2nd-order Taylor expansion	Affine mapping + 2nd-order correction.
Enhanced cond. expectation + 2nd order	Better conditional-Libor curve + 2nd-order correction.

Approach	**Core idea**
Shifted-lognormal moment matching + copula	
Moment matching method 1 **(selected)**	2-moment shifted-lognormal legs.
Moment matching method 3	3-moment marginal fit (adds skew).
Moment matching method 1 — fast (3 variants)	Same leg law; accelerated numerics.
Single-spread reduction	
Moment matching method 2	One shifted-DD spread (variance + skew); quadrature.
Kirk / Covariance approach	
Kirk based (variants 1, 2)	Kirk bivariate-lognormal price; v2 convexity-adjusted.
Hessian-based variance/covariance	Gradient + Hessian quadratic moments; no mapping.

Step 1 — Gaussian-copula spread price from the T-forward marginals.

- Baseline method. Couple the two swap-rate T-forward marginals Ψ_i^T by a Gaussian copula with term correlation ρ_{term}, $S_i(T) = (\Psi_i^T)^{-1}(\Phi(z_i))$, and integrate the spread payoff:

$$V_{\text{CMS}}(0) \approx \iint \Big((\Psi_1^T)^{-1}(\Phi(z_1)) - (\Psi_2^T)^{-1}(\Phi(z_2)) - K \Big)^+ \phi(z_1, z_2; \rho_{\text{term}})\, dz_1\, dz_2 .$$

- Each marginal is rebuilt from the annuity-measure law of S_i (closed form for the DD swap rate, density ψ^A) by the $\mathbb{Q}^A \to \mathbb{Q}^T$ change of measure $\frac{d\mathbb{Q}^T}{d\mathbb{Q}^A}\big|_T = \frac{A(0)}{P(0,T)\,A(T)}$, conditioned on $S(T)$:

$$\Psi^T(s) = \frac{A(0)}{P(0,T)} \int_{-\infty}^{s} \alpha(u)\, \psi^A(u)\, du, \qquad \alpha(s) = \mathbb{E}^A\Big[\tfrac{1}{A(T)} \,\Big|\, S(T) = s \Big].$$

- The whole method thus reduces to the *annuity-mapping function* $\alpha(s)$: the CMS convexity of each leg is carried by the reweighting $\Psi^T \neq \psi^A$ (Step 2).

Step 2 — The annuity mapping $\alpha(s)$ and its arbitrage correction.

- $1/A(T)$ is a function of the terminal LIBORs, $\frac{1}{A(T)} = \varrho(\mathbf{L}(T))$ with $\varrho(x) = \big(\sum_n \tau_n \prod_{i\le n}(1+\tau_i x_i)^{-1}\big)^{-1}$. Freeze the conditional law (Jensen — move $\mathbb{E}^A[\cdot \mid S]$ inside ϱ):

$$\alpha(s) = \mathbb{E}^A[\varrho(\mathbf{L}(T)) \mid S(T) = s] \approx \varrho\big(\mathbb{E}^A[\mathbf{L}(T) \mid S(T) = s]\big),$$

- with the conditional mean supplied by the same Gaussian regression as the Markovian-projection annex,

$$\mathbb{E}^A[L_n(T) \mid S(T) = s] \approx L_n(0) + \xi_n(T)\,(s - S(0)), \qquad \xi_n(T) = \frac{\int_0^T \lambda_n^\top \lambda_S \, dt}{\int_0^T \|\lambda_S\|^2 \, dt}.$$

- **Arbitrage correction.** No-arbitrage forces $\mathbb{E}^A[\alpha(S(T))] = \mathbb{E}^A[1/A(T)] = P(0,T)/A(0)$; the freezing violates it, so rescale $\tilde{\alpha} = \frac{P(0,T)}{A(0)}\,\alpha/\mathbb{E}^A[\alpha(S(T))]$. This baseline underlies the "limits of the baseline" slides; the retained approach (next) replaces the copula/annuity-mapping by *exact* $\mathbb{Q}^T$ moments.

Step 1 — Exact one-factor dynamics and Gyöngy projection.

- $S(t) = S(t, \mathbf{L}(t))$ is a smooth function of the DD forwards and a $\mathbb{Q}^A$-martingale, so by Itô the drift vanishes and only the diffusion survives:

$$dS(t) = \lambda_S(t, \mathbf{L}(t))^\top dW^A(t), \qquad \lambda_S(t, \mathbf{L}) = \sum_{n=j}^{k-1} \frac{\partial S}{\partial L_n}(t, \mathbf{L})\, \varphi_n(t, L_n)\, \lambda_n(t).$$

- Projecting on $dY^A = \lambda_S^\top dW^A / \|\lambda_S\|$ (a scalar $\mathbb{Q}^A$-BM, by Lévy) gives the *exact* one-factor form $dS(t) = \|\lambda_S(t, \mathbf{L}(t))\|\, dY^A(t)$ — non-Markovian, since the coefficient depends on all of $\mathbf{L}(t)$.
- **Gyöngy (1986):** S shares its marginals with the local-vol diffusion $d\hat{S} = \eta(t, \hat{S})\, d\hat{Y}$, where $\eta(t, x)^2 = \mathbb{E}^A\left[\|\lambda_S(t, \mathbf{L}(t))\|^2 \mid S(t) = x\right]$ — exact for swaption (marginal) pricing.

Step 2 — Freezing, Gaussian regression, displaced-diffusion fit.

- Freeze the state, $\eta(t,x)^2 \approx \|\lambda_S(t,\ell(t,x))\|^2$ with $\ell_n(t,x) = \mathbb{E}^A[L_n(t) \mid S(t) = x]$, and linearise by Gaussian regression of the forwards on S:

$$\ell_n(t,x) \approx \mathbb{E}^A[L_n(t)] + \xi_n(t)\,(x - S(0)),$$

$$\xi_n(t) = \frac{\mathrm{Cov}^A(L_n(t), S(t))}{\mathrm{Var}^A(S(t))} \approx \frac{\int_0^t \lambda_n^\top \lambda_S(u, \mathbf{L}(0))\,du}{\int_0^t \|\lambda_S(u, \mathbf{L}(0))\|^2\,du}.$$

- Fit the displaced diffusion $\eta(t,x) = p_S(t)\big(b_S(t)x + (1 - b_S(t))S(0)\big)$ by matching level and slope at the forward $x = S(0)$:

$$p_S(t) = \frac{\|\lambda_S(t, \mathbb{E}^A[\mathbf{L}])\|}{S(0)},$$

$$b_S(t) = \frac{1}{S(0)\,p_S(t)} \sum_{n=j}^{k-1} \frac{\partial \|\lambda_S(t, \mathbb{E}^A[\mathbf{L}])\|}{\partial L_n}\,\xi_n(t).$$

- **Time-homogenise (Piterbarg):** with $\nu(t)^2 = \int_0^t p_S^2$, the model-implied normal vol and skew are the variance-weighted averages $\bar\sigma_S^{\mathrm{DLN}} = (\frac{1}{T}\int_0^T p_S^2\,dt)^{1/2}$, $\bar b_S = S(0)\int_0^T b_S\,w_T\,dt$, $w_T = \nu^2 p_S^2 / \int_0^T \nu^2 p_S^2$.

Step 3 — The annuity-measure mean $\mathbb{E}^A[L_n(t)]$ (convexity).

- Step 2 freezes the forwards at their annuity-measure means $\mathbb{E}^A[L_n(t)]$, but L_n is *not* a $\mathbb{Q}^A$-martingale. It *is* driftless under its own forward measure $\mathbb{Q}^{n+1}$ (numeraire $P(\cdot, T_{n+1})$), so change measure with the numeraire ratio $Q_n(t) = A(t)/P(t, T_{n+1})$:

$$\mathbb{E}^A\big[L_n(t)\big] = \frac{1}{Q_n(0)}\,\mathbb{E}^{n+1}\big[Q_n(t)\,L_n(t)\big], \qquad \left.\frac{d\mathbb{Q}^A}{d\mathbb{Q}^{n+1}}\right|_t = \frac{Q_n(t)}{Q_n(0)}.$$

- Under $\mathbb{Q}^{n+1}$, $\mathbb{E}^{n+1}[L_n(t)] = L_n(0)$. Linearise the ratio $Q_n(\mathbf{L}(t))$ about $\mathbf{L}(0)$ and keep the leading frozen-coefficient covariance $\mathbb{E}^{n+1}[(L_i(t) - L_i(0))(L_n(t) - L_n(0))] \approx \int_0^t \lambda_i^\top \lambda_n\, ds$:

$$\mathbb{E}^A[L_n(t)] \approx L_n(0)\big(1 + c_n(t)\big), \qquad c_n(t) = \frac{1}{L_n(0)\,Q_n(0)} \sum_{i=j}^{k-1} \frac{\partial Q_n(0)}{\partial L_i(0)} \int_0^t \lambda_i(s)^\top \lambda_n(s)\, ds.$$

- $c_n(t)$ is the LMM convexity of the forward under the annuity measure; it supplies the intercept $\mathbb{E}^A[\mathbf{L}(t)]$ of the Gaussian regression of Step 2 — the freezing point at which $p_S(t)$ and the skew $b_S(t)$ are evaluated.

Volatility and covariance averaging is *exact*.

- With the displacement held constant, each leg $X_i = S_i + \delta_i$ is a driftless martingale with *deterministic* log-vol, $dX_i = \lambda_i(t)\, X_i\, dW_i^{A_i}$, $\lambda_i(t) = b_{S_i}^{\text{eff}} \sigma_{S_i}(t)$. Its log-return $M_t = \int_0^t \lambda_i\, dW^{A_i}$ is a continuous martingale with *deterministic* clock

$$\langle M \rangle_t = \int_0^t \lambda_i^2(u)\, du =: V_i(t).$$

- **Dambis–Dubins–Schwarz** (Karatzas–Shreve, Thm. 3.4.6 [15]): a continuous martingale is a Brownian motion run on its own variance clock, $M_t = B_{V_i(t)}$. Therefore

$$X_i(T) = X_i(0)\, e^{B_{V_i(T)} - \frac{1}{2} V_i(T)} \overset{d}{=} X_i(0)\, e^{\bar{\lambda}_i \sqrt{T}\, Z - \frac{1}{2} \bar{\lambda}_i^2 T}, \qquad \bar{\lambda}_i^2 = \frac{1}{T} \int_0^T \lambda_i^2\, dt, \quad Z \sim \mathcal{N}(0,1).$$

- The terminal law — hence *every* European price on the leg — depends on $\lambda_i(\cdot)$ only through $V_i(T)$, so the constant-parameter leg reproduces it *exactly*; with $\lambda_i = b_{S_i}^{\text{eff}} \sigma_{S_i}$ this is precisely $(\sigma_{S_i}^{\text{eff}})^2 = \frac{1}{T} \int_0^T \sigma_{S_i}^2\, dt$. Jointly, the deterministic cross-variation $\langle M_1, M_2 \rangle_T = \int_0^T \rho\, \sigma_{S_1} \sigma_{S_2}\, dt$ is reproduced by ρ^{eff}, matching the terminal Gaussian copula exactly.

Displacement (skew) averaging is an *approximation*.

- The exactness above needed a *constant* displacement: only then is $X_i = S_i + \delta_i$ a time-changed geometric Brownian motion. A genuinely time-dependent $b_{S_i}(t)$ makes $\delta_i(t)$ move, so the local-vol SDE $dS_i = \varphi_i(S_i, b_{S_i}(t))\,\sigma_{S_i}(t)\,dW$ is no longer a clean time change and no exact time-change reduction is available.
- Piterbarg's $b_{S_i}^{\text{eff}}$ instead matches the *effective skew* to leading order in a small-volatility expansion, via the variance-weighted average

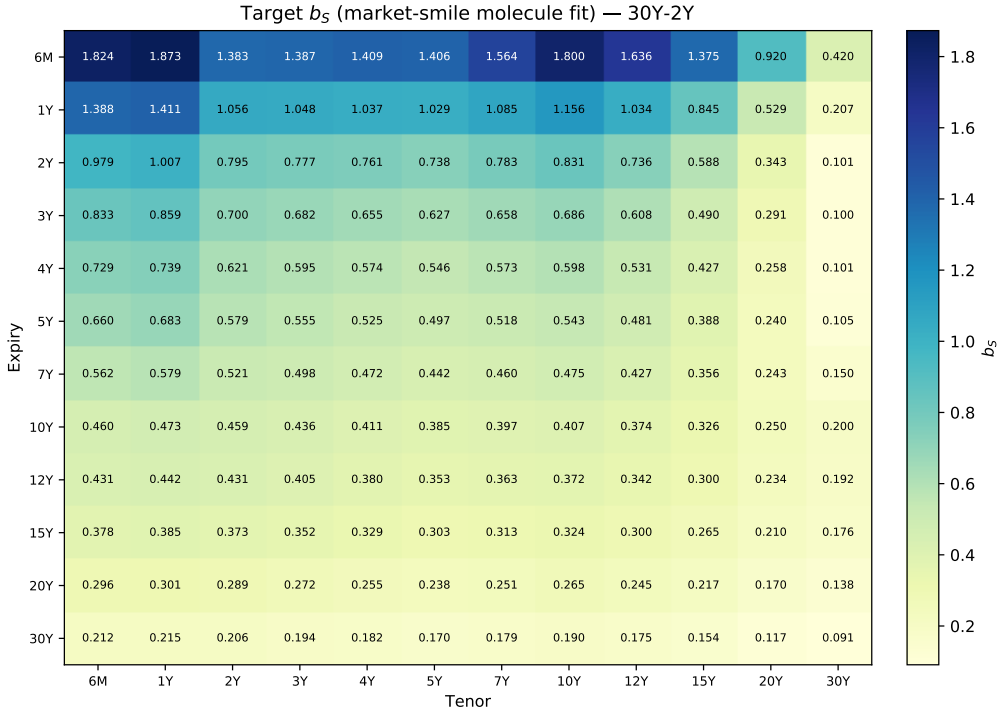
$$b_{S_i}^{\text{eff}} = \frac{\int_0^T b_{S_i}(t)\,\sigma_{S_i}^2(t)\,\omega_i(t)\,dt}{\int_0^T \sigma_{S_i}^2(t)\,\omega_i(t)\,dt}, \qquad \omega_i(t) = \int_0^t \sigma_{S_i}^2(u)\,du,$$

- Complete 10Y-2Y CMS spread fit (ATM normal vols, bps): Market, Model (Monte Carlo), closed-form Approx., and residuals.

Expiry	Market	Model	Approx.	Model−Mkt	Model−Apx
6M	41.21	41.54	41.60	+0.34	−0.06
1Y	41.64	41.06	41.22	−0.58	−0.16
2Y	39.56	39.27	39.48	−0.30	−0.21
3Y	37.67	37.83	37.76	+0.16	+0.07
4Y	36.33	36.29	36.29	−0.04	0.00
5Y	35.20	35.11	35.15	−0.09	−0.05
6Y	34.27	34.18	34.24	−0.09	−0.06
7Y	33.46	33.49	33.49	+0.03	0.00
8Y	32.69	32.55	32.74	−0.14	−0.19
9Y	32.05	31.98	32.18	−0.07	−0.20
10Y	31.63	31.39	31.55	−0.24	−0.16
12Y	30.74	30.74	30.71	0.00	+0.03
15Y	29.78	29.88	29.83	+0.10	+0.06
20Y	28.90	28.97	28.91	+0.07	+0.06
25Y	28.39	28.20	28.40	−0.19	−0.20
30Y	28.10	27.65	28.07	−0.46	−0.42

- Complete 30Y-2Y CMS spread fit (ATM normal vols, bps).

Expiry	Market	Model	Approx.	Model−Mkt	Model−Apx
6M	61.02	62.21	61.37	+1.19	+0.84
1Y	60.44	60.78	60.16	+0.34	+0.62
2Y	57.12	57.03	56.97	−0.10	+0.06
3Y	54.24	54.69	54.28	+0.45	+0.41
4Y	52.14	52.36	52.12	+0.22	+0.25
5Y	50.30	50.58	50.32	+0.28	+0.26
6Y	48.81	48.66	48.83	−0.15	−0.17
7Y	47.51	46.95	47.53	−0.55	−0.57
8Y	46.31	45.61	46.32	−0.70	−0.71
9Y	45.34	44.71	45.43	−0.63	−0.73
10Y	44.63	43.86	44.55	−0.76	−0.68
12Y	43.21	42.33	43.20	−0.88	−0.87
15Y	41.72	40.98	41.71	−0.74	−0.73
20Y	40.22	39.57	40.28	−0.65	−0.71
25Y	40.33	39.02	40.34	−1.31	−1.32
30Y	40.67	38.92	40.61	−1.75	−1.69



- Market-implied displacement $\hat{b}_S$ surface for the 30Y-2Y basket, from the per-swaption optimisation — the displacement target fed into the displaced-lognormal calibration.

- **Origins of the LMM:**
  - Short-rate models (Vasicek [33], CIR [6], Hull–White [12], Black–Karasinski [3]): single-factor, cannot capture term structure decorrelation.
  - HJM framework [9]: models the full forward curve, but state variables are unobservable instantaneous rates.
  - LMM (Brace–Gatarek–Musiela [4]; Jamshidian [14]): directly models observable forward LIBOR rates; recovers Black/Bachelier caplet formulas by construction.
- **Calibration strategies:**
  - Local calibration: product-specific instrument subset [26].
  - Global calibration: broad grid of caps and swaptions; made tractable by Rebonato [28] and Andersen–Piterbarg [2] via analytical approximations and parsimonious parametrizations.
- **Volatility & correlation specifications:**
  - Parametric: Rebonato *abcd* form [27] — parsimonious but limited flexibility.
  - Grid-based (Pedersen [25], Sidenius [32], Andersen–Piterbarg [2]): industry standard; requires regularization.
  - Correlation: exponential decorrelation with maturity-dependent speed and floor [30, 31].
- **Optimization:** Levenberg–Marquardt [24] exploits least-squares structure; outperforms general-purpose quasi-Newton (L-BFGS [18]) for this problem class.

- **Monte Carlo simulation & American Monte Carlo:**
  - Discretisation: predictor-corrector scheme [13] for accurate drift approximation.
  - Variance reduction: antithetic variates, control variates [7].
  - Early exercise: Longstaff–Schwartz regression [19] for Bermudan swaptions under the LMM [1].
- **Spread options & correlation calibration:**
  - Yield curve spread caps provide direct market-implied information on inter-rate correlations.
  - Kirk approximation [16]; LMM-consistent spread option formulas [2].
- **Factor reduction:**
  - PCA: 3 factors (level, slope, curvature) explain $>$95% of variance [17].
- **Extensions & related developments:**
  - Local/stochastic volatility: CEV, displaced diffusion, SABR-LMM [8, 23, 29].
  - Forward Market Model for RFR rates [20].
  - Multicurve framework [10, 22].
  - Calibration of volatility models through deep learning [5, 11]

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